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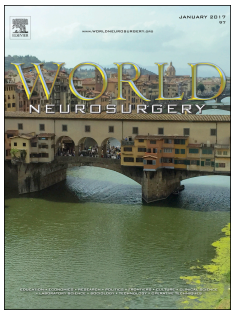

Same-Year Repeat Concussions in the National Football League: Trends from 2015 through 2019

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Title: Same-Year Repeat Concussions in the National Football League: Trends from 2015 through 2019

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Abstract:

Background: Sport-related concussion (SRC) prevention and management is a focus of the National Football League (NFL). While most prior reports evaluate SRC incidence, few have studied same-year repeat concussions. The objective of this study was to evaluate the frequency of same-year repeat concussions in the NFL.

Methods: A retrospective, case-control study of same-year repeat concussion in the NFL from 2015-2019 was performed. The electronic health record of the NFL was reviewed for players sustaining concussions and same-year repeat concussions. Wilcoxon rank-sum tests were used to calculate same-year repeat concussion rates, and risk ratios and 95% confidence intervals (CIs) were estimated using log-binomial regression.

Results: From 2015-2019, the risk of sustaining a same-year repeat concussion in the NFL ranged from 0.38% to 0.69% per season. Among players who sustained a concussion, the risk of a same-year repeat concussion ranged from 5.3% to 8.3%, which did not differ significantly from the risk of sustaining a single concussion, which ranged from 6.2% to 8.3%. There was a median of 38 participation days between initial and same-year repeat concussion. Players missed more time from same-year repeat concussion (median 12 days) compared to both single (median 9 days; $p<0.0001$) and initial (median 9.5 days, $p=0.002$) concussions.

Conclusions: The risk of a repeat concussion was similar to the risk of sustaining a single concussion among NFL players. More time was missed following a same-year repeat concussion compared to single or initial concussions. Further research is needed to maximize player safety and minimize same-year repeat concussions.

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Introduction

Increasing knowledge and awareness surrounding sport-related concussion (SRC) in recent decades has resulted in concerted efforts directed toward reducing repeat SRC rates. Repeat or recurrent concussions have been associated with a slower recovery,[1] depression,[2] sleep disturbances,[3] late-life cognitive impairment,[4] and longer return-to-play (RTP) times compared to first-time injuries.[5,6,7] Though many studies have evaluated the risk of repeat SRC throughout an athlete's career,[2,4,8–10] few have delved into the incidence of same-year repeat concussions, especially among elite American football players.

A previous study reviewed concussions in the National Football League (NFL) between 1996 and 2001 and reported a 5.5% rate of same-season repeat concussions.[11] A follow-up study found a similar risk of repeat concussions between two time periods of the 1996-2001 vs. 2002-2007 seasons (24.7% vs. 24.2%).[11] Although the second follow-up study did not restrict the definition of repeat concussion to same-year repeat concussion, the authors found that 90% of repeat concussions occurred more than 28 days after the initial concussion. The average time between concussions was 456 days, signifying most injuries likely occurred in different seasons.[11]

Several evidence-based changes in concussion detection and RTP protocols have been implemented in the NFL in recent years.[12–14] With respect to RTP, players must pass a series of clinical qualifications to be cleared for play. From 2015-2019, between 283-302 concussions were reported per year (offseason through the postseason) in the NFL.[15] The most recent publication regarding same-year repeat concussion in the NFL is from 1996-2001;[16] more recent data is needed to fully understand the current scope of same-year repeat concussion.

In the current study, we sought to: 1) report the risk of same-year repeat concussions in the NFL, 2) describe the time pattern of when same-year repeat concussions occur, and 3) present RTP time and intervening time between same-year repeat concussions. Given recent evidence-based changes in concussion management, the staged approach for RTP, and plateaued rates of SRC due to increased awareness and prevention measures,[15] we would expect the risk of same-year repeat concussion in the NFL to have plateaued from previous studies.

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Methods

Research Design

A retrospective, observational cohort study of NFL players during the 2015 through 2019 seasons was conducted in accordance with the NFL and National Football League Players Association (NFLPA) Medical and Scientific Research Protocol Approval Process[17] and was approved by the [*blinded for review*] institutional review board (0006018120). Medical records of all players with football-related concussions were reviewed for this study.

Participants

All players who sustained at least one concussion during the 2015 through 2019 NFL seasons were included. All NFL-related concussions were included, regardless of severity, timing within the season, or activity when the injury occurred. Player position was categorized by position at time of concussion, with all players injured on special teams plays categorized as a special teams player, rather than by specific roster position: offensive line, running back, tight end, quarterback, wide receiver, defensive secondary, defensive line, linebacker, or special teams (e.g. kicker, punter, long snapper).

Data Collection

A player sustaining a *same-year repeat concussion* was defined as a player with a second concussion at any point during the same NFL year. An "NFL year" runs from mid-March through mid-March and is denoted by the initial year. The year begins with an offseason program and progresses in order of preseason, regular season, and postseason. An *initial concussion* was defined as the first concussion in a player who sustained a same-year repeat concussion. A player with a *single concussion* was defined as a player who had only one concussion during the full NFL year (no subsequent concussions in that same year).

Several time points were considered when evaluating RTP and intervening time between SRCs. RTP was defined as the time when the athlete had fully cleared concussion protocol and was released to full-contact practice and games. Any limitations placed on a player meant they were still completing the concussion protocol and had not yet returned. *Missed time* was the number of days between removal from participation due to the concussion and return to full participation in football activities, exclusive of removal and return dates. *Participation days* was

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defined as the number of intervening days between full RTP to football activities and the onset date of the second concussion for players with a same-year repeat concussion, inclusive of both the return date and the onset date of the second concussion. Additional concussions after the second were not included when calculating missed time. A schematic depicting key timepoints among players with same-year repeat concussions is seen in **Figure 1.**

Statistical Analysis

Continuous data are presented as median, range, and interquartile range due to skewness of days missed. Wilcoxon rank-sum tests were used to compare medians. Count data are presented as n (%). One-season risk was calculated in two ways: 1) the risk of a concussion or same-year repeat concussion among all players in the NFL, which was calculated as the number of players with a concussion divided by the total number of players under contract each full NFL season, and 2) the risk of a same-year repeat concussion among players who had at least one concussion. Risk ratios and 95% confidence intervals (CIs) were estimated using log-binomial regression. Significance was considered at alpha of 0.05. All statistical analyses were performed in SAS 9.4 (SAS Institute, Cary, NC).

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Results

Risk of Same-Year Repeat Concussions

The total number of concussions and same-year repeat concussions is presented in **Table 1.** Among all players, the risk of sustaining a same-year repeat concussion ranged from 0.38% in 2019 to 0.69% in 2017. For those sustaining at least one concussion during an NFL year, the risk of same-year repeat concussion ranged from 5.3% (2019) to 8.3% (2017). This was similar to the risk of sustaining any concussion, which ranged from 6.2% (2018) to 8.3% (2017).

Characteristics of Players with Same-Year Repeat Concussions

Rates of same-year repeat concussions by position at time of concussion were similar **(Table 2).** Notable trends include players on special teams plays and in the defensive secondary, who sustained the largest proportions of both repeat and single concussions. Percentages comparing single concussions to same-year repeat concussions differed among: 1) tight ends who sustained 7.9% of single concussions but 12.0% of repeat concussions, 2) wide receivers

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who sustained 12.0% of single concussions but 16.9% of repeat concussions, and 3) offensive linemen who sustained 17.8% of single concussions but 10.8% of repeat concussions.

Timing of Same-Year Repeat Concussions

Among 83 players with same-year repeat concussions, 48% (n=40) experienced the first concussion during the offseason or preseason, and 82% (n=68) had concussions that occurred during at least one preseason and one regular season or two regular season injuries (**Figure 2**).

Participation Days Between Injuries

Players participated a median of 38 days between the first and same-year repeat concussions **(Table 3).** The minimum was two days, and maximum was 150 days. In 18.8% of players sustaining a same-year repeat concussion, the second concussion occurred within two weeks of return from the initial concussion. There were three players who sustained a same-year repeat concussion two participation days after returning from the initial concussion. Among those three cases, the missed time for the initial concussion ranged from 11-19 days, and the same-year repeat concussions occurred during a regular season game (n=2) or preseason practice (n=1). Out of 83 players with a same-year repeat concussion, 55 (66.3%) sustained the second concussion 36 or more days after the initial concussion; 18 (21.7%) sustained the second concussion 22-35 days after the initial concussion; and the remaining 10 (12.0%) sustained the concussion 14-21 days after the initial concussion.

Return-to-Play

Missed time due to first and same-year repeat concussion is seen in **Table 4.** Missed time following single concussions (median 9 days) was similar to missed time following initial concussions (median 9.5 days) (p=0.20); however, missed time after a same-year repeat concussion was longer (median 12 days) than both single (p<0.0001) and initial (p=0.002) concussions**.** Players who missed 8 or more days following their initial concussion trended toward a higher risk of same-year repeat concussion compared to players who missed 7 or fewer days from the initial concussion (8.5% vs. 4.9%; RR=1.48, 95% CI 0.94, 2.31, p=0.09).

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Discussion

This study aimed to summarize the risk of same-year repeat concussion in NFL players over 5 seasons. From 2015-2019, the risk of any NFL player sustaining a same-year repeat concussion ranged from 0.38% to 0.69%. In players sustaining at least one concussion, the risk of sustaining a same-year repeat concussion was 5.3% to 8.3%, which was similar to the risk of sustaining at least one concussion at 6.2% to 8.0%. Time missed for the same-year repeat concussion was longer than for a single concussion (12 vs. 9 days). A trend toward higher risk of same-year repeat concussion was seen in players who missed 8 or more days after the initial concussion, though this trend was not significant.

Few studies have evaluated the risk of same-year repeat concussion at any level, and only two studies at the NFL level exist. From 1996-2001, Pellman et al.[16] reported that of 650 players who sustained a concussion, 36 (5.5%) sustained at least one same-season repeat concussion, which is similar to the 7.0% reported in this study. One explanation for the small difference is the inclusion of the off-season in our study, which accounted for two of the initial concussions in the same-year concussion group. Unfortunately, Pellman et al. did not report the risk of same-season repeat concussion for all NFL players in that time period, which in the current study was low at 0.51%.[16] Casson, et al.[11] compared the data from the 1996-2001 and 2001-2006 seasons and found that the risk of repeat concussions – more than one concussion in the study window and not same-season repeat concussion – remained similar across the two time periods (24.7% vs. 24.2%, respectively). While that study did not specifically report same-season repeat concussions, it did note that the majority of players (89.8%) had more than 28 days between concussions (average 456 days), with only 2.8% experiencing a repeat concussion within the same week.[11]

Since the publication of these early studies, and prior to the initial years of the current study, several policy changes have been implemented regarding the management of concussed players in the NFL, including indicators for immediate removal from play, a stepwise RTP protocol, and mandatory evaluation by an independent neurological consultant.[14] In the current study, no player sustained a repeat concussion within 13 days of onset of the initial concussion. Mack, et al., found that the concussion risk declined significantly in the 2018-2019 seasons (6.2% and 7.1%, respectively) compared to the 2015-2017 seasons (7.4%-8.3%), thought to be attributed to a dedicated focus on concussion awareness prior to the 2018 season.[15] Additional

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rule changes after 2018 included elimination of blind-side blocks and updated kickoff rules. Helmets were also tested in the laboratory,[18] the results of which were made available to players and teams. A subsequent policy mandated that players choose among higher performing helmets, with poorer performing helmets being banned. Our data on same-year repeat concussions in the same time interval do not demonstrate a significant decrease in the rate of same-year repeat concussions, though 2019 did have the lowest rate (0.38%). This is likely due to the relatively low numbers of same-year repeat concussions and a decrease in the denominator of overall concussions.

Same-year repeat concussions at the collegiate level also warrant discussion in the context of elite American football. The National Collegiate Athletic Association (NCAA) Concussion study (1999-2001) by Guskiewicz, et al.[1] found a 6.5% risk of same-season repeat concussion among football players. In these athletes, the median RTP was 3.0 days and 91% of repeat concussions occurred within 10 days of the initial concussion. A subsequent study involving the Concussion Assessment, Research, and Education (CARE) Consortium, collecting data from 2014-2017,[6] found a same-season repeat concussion rate of 3.9%, demonstrating a significant decline from the initial NCAA Concussion Study. RTP time was longer in the CARE Consortium compared to the 1999-2001 study (median 12.2 days) and only 3.7% of same-season repeat concussions occurred within 10 days of the initial concussion.[6] Though comparisons between playing levels are made cautiously, the current NFL data demonstrate a higher risk of same-year season repeat concussion (6.8%) in NFL players compared to the CARE consortium cohort of collegiate football players (3.9%).[6,19] One explanation for higher risk of same-year repeat concussion may be differences in the length of the study period. The current study included the full NFL year (offseason, 4 preseason games, 16 regular season games, and potential for 4 postseason games), which is longer than a NCAA season. The NCAA football season typically runs from the beginning of workouts in July through late December or early January bowl games, with between 12 and 1 games for most teams. Other explanations may include differences in the style of play, access and availability of medical care, motivations for RTP, differences in data collection, or other differences between collegiate and professional football settings.

The current study suggests that the risk of sustaining a same-year repeat concussion is similar to the risk of sustaining a single concussion. Stable rates of same-year repeat and single

concussion are aligned with the likelihood that players have fully recovered by the time they return from their injuries and re-enter the population at risk with a similar risk of sustaining a concussion, rather than having an elevated risk for a repeat concussion. Hypothetically, if an elevated risk of same-year repeat concussion compared to single concussions were to be found, this may indicate a potential premature RTP following the initial concussion. Premature RTP has been associated with a risk of additional concussion and a prolonged recovery,[1] in addition to an increased risk of musculoskeletal injury,[20–22] or even second impact syndrome.[23,24]

RTP times in our cohort were longer with repeat concussions (median 12 days) compared to single concussions (median 9 days), which is also consistent with several previous studies demonstrating prolonged recovery time from repeat concussions in high-school,[25,26] collegiate,[1,6] and professional[27] sports. Though the current study cannot assess causation, these data may indicate that a second concussion within the same season may result in an injury requiring more recovery time. An alternative possibility is that team physicians and athletic trainers tend to be more cautious when athletes incur a second concussion in the same season. Both possibilities require further empirical study.

Among players with at least one concussion in a season, those who returned to play after eight or more days were more likely to sustain a same-year repeat concussion than those who returned to play after seven or fewer days, though this trend was not significant.[26,27] The data suggest that a concussion resulting in longer RTP , likely representing a more severe concussion, may place players at higher risk for sustaining a same-year repeat concussion; players with 8 or more days missed after an initial concussion had an estimated 48% higher risk of a same-year repeat concussions than players with 7 or fewer days missed. Identifying additional risk factors for repeat concussion may be able to help reduce the rate of repeat concussions.

This study has limitations. This study focused on two or more concussions within an NFL season; it did not evaluate concussions across multiple seasons or incorporate head injuries prior to joining the NFL. It is possible that players had sustained one or more concussions prior to the 2015 season in or outside of the NFL, or sustained concussions across multiple seasons within the study. Additionally, same-year repeat concussions may have been impacted by the timing of the initial concussion within the season. For example, if a player sustained a concussion in the latter portion of the year, they would have less time to sustain a same-year repeat concussion, as there would be fewer opportunities to do so with the time remaining in the season. Conversely, a

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player who sustained a concussion at the beginning of the year may have more opportunities to sustain a same-year repeat concussion. Similarly, time to return to participation may have been influenced by timing in the season and other factors. Notably, this analysis does not account for granular time at risk for injury (e.g. player-games or player-days) as participation data outside of games in limited. Also, our data do not account for limitations (e.g. play count or rep maximum) placed on players by coaches, trainers, or athletes themselves once they are cleared for full participation. However, investigating each NFL year as a unique interval allows for the study of same-year risk for repeat concussion, which has not been reported since the newest iteration of the NFL concussion protocols. The small number of same-year repeat concussions limits the ability to perform subgroup analyses for timing of concussions during the year, though this may be a topic for future study. This study benefits from the inclusion of all players within a single league, and consistency of data collection and robust, uniform detection procedures[12,13,28] across several seasons to minimize bias.

Conclusions

This study investigates the risk of same-year repeat concussion in the NFL over 5 seasons from 2015 through 2019. Repeat concussions occurred at a similar rate to initial concussions, indicating that players may not be at increased risk for a subsequent concussion when following evidence-informed RTP protocols. More missed time from an initial concussion was associated with an increased risk of repeat concussion, suggesting that a more severe concussion may be associated with an increased risk of subsequent concussion. This study highlights the effectiveness of a robust concussion identification system and RTP protocol. Physicians involved in the evaluation and management of concussion should continue to pay significant attention to the systems of identification of concussion, as well as their RTP protocols. Future research should address risk factors for sustaining repeat concussions in order to enhance player safety.

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Figure Legend

Figure 1. Time interval of initial and same-year repeat concussion.

Figure 2. Distribution of same-year repeat concussions by season, National Football League 2015-2019 (n=83 concussions)

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Table 1. Risk of same-year repeat concussion, National Football League 2015-2019 (n=1,302 concussions)

	Concussed players	Players with same-year repeat concussion	Risk of any concussion*	Risk of same-year repeat concussion*	Risk of same-year repeat concussion among concussed players
2015	263	19	8.0%	0.58%	7.2%
2016	242	13	7.4%	0.40%	5.4%
2017	277	23	8.3%	0.69%	8.3%
2018	206	16	6.2%	0.48%	7.8%
2019	226	12	7.1%	0.38%	5.3%

* All unique player signings within each NFL season from 2015-2019 were included in the denominator for the risk calculations (2015=3,278; 2016=3,288; 2017=3,337; 2018=3,333; 2019=3,167).

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Table 2. Distribution of player position at time of concussion, National Football League 2015-2019 (n=1,302 concussions)

	Single Concussions*		Same-Year Repeat Concussions†			
			Initial Concussion		Second Concussion	
	N	Column %	N	Column %	N	Column %
Offensive Line	200	17.8%	9	10.8%	9	10.8%
Running Back	71	6.3%	5	6.0%	3	3.6%
Tight End	89	7.9%	9	10.8%	10	12.0%
Quarterback	37	3.3%	4	4.8%	4	4.8%
Wide Receiver	135	12.0%	12	14.5%	14	16.9%
Defensive Secondary	233	20.7%	16	19.3%	14	16.9%
Defensive Line	88	7.8%	5	6.0%	5	6.0%
Linebacker	101	9.0%	6	7.2%	9	10.8%
Special Teams	172	15.3%	17	20.5%	15	18.1%

* Position is unknown for n=5 single concussions.
† Players could have been playing different positions at the time of initial vs. second concussions; n=5 third same-year repeat concussions are not shown.

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Table 3. Distribution of participation days prior to same-year repeat concussion, National Football League 2015-2019 (n=83 concussions)

	Number of same-year repeat concussions*	Median	Range	IQR
Overall	80	38	2-150	19-73.5
2015	17	31	2-143	20-56
2016	12	23	7-107	17-54.5
2017	23	68	2-142	35-88
2018	16	22.5	2-150	5.5-63
2019	12	29.5	11-104	19-69

IQR: Interquartile range
*N=3 same-season repeat concussions were excluded since the corresponding first concussion had a missing or invalid return date [2015 (n=2) and 2016 (n=1)].

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Table 4. Distribution of missed time due to concussions, National Football League 2015-2019 (n=1,302 concussions)

	Single Concussions*				Same-Year Repeat Concussions†							
					Initial Concussion				Second Concussion			
	N	Median	Range	IQR	N	Median	Range	IQR	N	Median	Range	IQR
2015	210	9	0-157	5, 15	17	9	3-23	5, 17	18	14	0-181	6, 38
2016	203	9	1-155	5, 12	12	10	4-37	7.5, 13.5	11	9	2-49	4, 18
2017	235	9	0-168	5, 13	23	8	2-22	4, 11	22	12.5	2-141	8, 47
2018	178	8	0-318	5, 12	16	11.5	3-30	7.5, 18	15	49	3-79	7, 72
2019	198	9	0-224	5, 14	12	10	4-18	8, 14	11	18	8-93	10, 42

IQR: Interquartile range
* Single concussions with missing or invalid return dates are excluded (n=34 in 2015, n=26 in 2016, n=19 in 2017, n=12 in 2018, and n=16 in 2019).
† Concussions with missing or invalid return dates for initial concussion (n=2 in 2015 and n=1 in 2016) and second concussion are excluded (n=1 in 2015, n=2 in 2016, n=1 in 2017, n=1 in 2018, and n=1 in 2019); n=5 third same-year repeat concussions are not shown.

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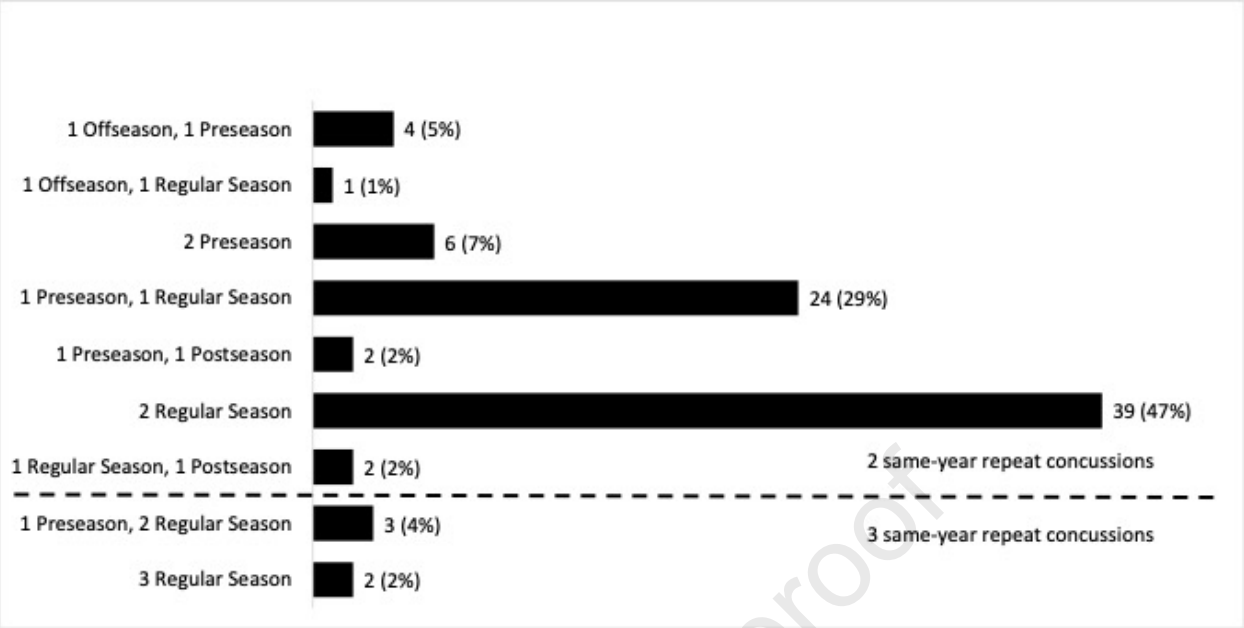

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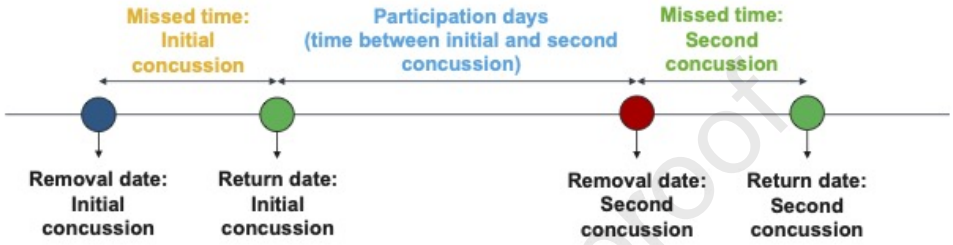

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Abbreviations: CARE, Concussion Assessment, Research, and Education; CI, Confidence Interval; NCAA, National Collegiate Athletic Association; NFL, National Football League; NFLPA, National Football League-Players Association; RTP, Return-to-play; SRC, Sport-related concussion

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